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RESEARCH ARTICLE

An Optimised Multishot EPI Protocol for Ex Vivo Mouse Brain Diffusion MRI With Gd-DOTA Contrast Enhancement

Elise Cosenza[1] | Nadège Corbin[2,3] | Emeline J. Ribot[2] | Arnaud Boré[3,4] | Maxime Descoteaux[3,4] | Sylvain Miraux[2,3,5] | Laurent Petit[1,3]

[1]Groupe d'Imagerie Neurofonctionnelle, Institut Des Maladies Neurodégénératives (GIN-IMN), UMR5293, CNRS, CEA, Université de Bordeaux, Bordeaux, France | [2]Centre de Résonance Magnétique des Systèmes Biologiques (CRMSB), UMR5536 CNRS, Université de Bordeaux, Bordeaux, France | [3]IRP/LIA OpTeam, CNRS Biologie et Université de Bordeaux, France–Université de Sherbrooke, Canada, Bordeaux, France | [4]Sherbrooke Connectivity Imaging Laboratory (SCIL), Department of Computer Science, Faculty of Sciences, Université de Sherbrooke, Sherbrooke, Quebec, Canada | [5]Plateforme d'Imagerie Biomédicale (pIBIO), UAR3767 CNRS, Université de Bordeaux, Bordeaux, France

Correspondence: Sylvain Miraux (sylvain.miraux@rmsb.u-bordeaux.fr) | Laurent Petit (laurent.petit@u-bordeaux.fr)

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ABSTRACT

Diffusion-weighted magnetic resonance imaging (dMRI) in small animal models is often limited by long acquisition times, low signal-to-noise ratio (SNR) and magnetic susceptibility artefacts. We developed and optimised a 3D spin-echo multishot EPI (SE-msEPI) protocol combined with Gd-DOTA administration to enable efficient, high-quality ex vivo diffusion imaging of the mouse brain. Systematic testing demonstrated that 12 EPI shots provided the optimal trade-off between artefact reduction, SNR and acquisition duration. Gd-DOTA incubation at 3 mM for 7 days allowed the TR to be reduced from 1500 to 400 ms without compromising diffusion metrics, yielding more than threefold SNR increase. This optimisation enabled whole-brain acquisitions at 100-µm isotropic resolution with 60 directions ($b = 3000\,\mathrm{s/mm^2}$) in ~11 h, a fourfold acceleration compared to standard long-TR protocols. Fractional anisotropy values in Gd-DOTA–enhanced acquisitions closely matched those of the long-TR reference, while low-SNR short-TR baseline and postwashout acquisitions exhibited systematic FA underestimation. Residual error analysis and tractography performance metrics further demonstrated the improved image quality and anatomically plausible reconstruction of white matter pathways under Gd-DOTA–enhanced conditions. Beyond these technical gains, the optimised protocol provides a reproducible framework for high-throughput ex vivo diffusion MRI, enabling connectomics and microstructural studies in mouse models with greater efficiency, increased reuse of tissue sample and reduced animal use.

Abbreviations: AC, anterior commissure; ACO, anterior commissure, olfactory part; AMBA, Allen mouse brain atlas; ANTS, advanced normalisation tools; CC, corpus callosum; CCB, corpus callosum body; DIPY, diffusion imaging in python; DIPY-based SCILPY, tools developed by the Sherbrooke connectivity imaging lab (SCIL); dMRI, diffusion magnetic resonance imaging; DTI, diffusion tensor imaging; DWI, diffusion-weighted imaging; DW-GRASE, diffusion-weighted gradient spin-echo; EPI, echo planar imaging; FA, fractional anisotropy; FAIR-RARE, Flow-sensitive alternating inversion recovery combined with rapid acquisition with relaxation enhancement; FSL, Oxford centre for functional MRI of the brain (FMRIB) software library; Gd-DOTA, gadolinium complex of 1,4,7,10-tetraazacyclododecane-1,4,7,10-tetraacetic acid; HARDI, high-angular-resolution diffusion imaging; INT, internal capsule; MESE, multiecho spin-echo; MRtrix3, magnetic resonance tractography; N4, nonparametric nonuniform intensity normalisation; ODF, orientation distribution function; PBS, phosphate buffer saline; PFA, paraformaldehyde; r_1, r_2, relaxivities; ROI, region of interest; SD, standard deviation; SNR, signal-to-noise ratio; SE-msEPI, spin-echo multishot echo planar imaging; SE-ssEPI, spin-echo single-shot echo planar imaging; WM, white matter.

1 | Introduction

Diffusion-weighted magnetic resonance imaging (dMRI) and tractography provide valuable insights into the organisation of white matter fibres, leading to a deeper understanding of brain connectivity [1]. While these methods are widely used to investigate human brain architecture and its relationship with brain function, their application to small animal models remains technically challenging [2]. In rodents, traditional microscopy-based techniques such as confocal microscopy or biophotonics provide very high spatial resolution and detailed cellular-level information. Although these invasive approaches are feasible in animal models and partially reduce the reliance on neuroimaging techniques, they are not suitable for longitudinal investigations. In this context, dMRI remains a valuable noninvasive tool for visualising and tracking brain connectivity over time. However, achieving sufficiently high spatial resolution ($\leq 100\,\mu$m) while maintaining an adequate signal-to-noise ratio (SNR) remains challenging in small animal imaging [3, 4].

Diffusion imaging and tractography require long acquisition times due to the small size of the rodent brain and the need for multiple diffusion-encoding directions, particularly for high angular resolution diffusion imaging. Ex vivo acquisitions are therefore often performed using 3D spin-echo-based sequences, which can provide high SNR and very high spatial resolution but may require scan times of several days [5, 6]. In fixed tissue, reduced diffusivity leads to decreased diffusion contrast, motivating the use of higher *b* values and long acquisitions. Moreover, isotropic spatial resolution is essential for accurate tractography reconstruction [3, 7]. Reducing acquisition time while preserving image quality thus remains a major challenge for ex vivo diffusion MRI.

Careful optimisation of the protocol is necessary to reduce acquisition time [8, 9], for example by using diffusion-weighted accelerated readout strategies such as DW-GRASE. Another way to shorten acquisition times, diffusion MRI is typically acquired in vivo using spin-echo echo-planar imaging (SE-EPI) readouts [10, 11]. However, echo-planar imaging exhibits higher sensitivity to B0 inhomogeneity than standard spin-echo-based sequences, resulting in geometric distortions and signal dropout. These effects are particularly pronounced in ex vivo samples, where air bubbles or susceptibility differences at tissue boundaries can exacerbate B0 inhomogeneity. Segmenting the k-space into multiple excitations (multishot EPI) reduces the echo train duration, which decreases the associated distortions and offers the possibility of reducing the TE. A lower TE helps to reduce signal loss, but an increase in the number of excitations results in a longer acquisition time. An alternative strategy to reduce scan duration is to shorten the TR; however, this typically lowers SNR unless tissue T_1 relaxation times are reduced.

The goal of this study was to develop and optimise a fast 3D multishot spin-echo EPI diffusion MRI protocol for the ex vivo mouse brain that balances acquisition time, image quality and robustness to artefacts [3, 12–14]. This optimisation focused on determining an appropriate number of shots and on refining sample preparation using a gadolinium-based contrast agent (Gd-DOTA) to shorten T_1 relaxation time constants and enable shorter TR without compromising SNR. Although gadolinium primarily acts as a T_1 contrast agent, excessive concentrations can also shorten T_2 and lead to signal loss [15], making careful optimisation of concentration necessary. Optimisation of Gd-DOTA concentration has previously been demonstrated in marmoset and rat studies [12, 16].

Here, we investigate an immersion-based Gd-DOTA preparation strategy for ex vivo mouse brain diffusion MRI, building on previous immersion approaches [8, 17], with a specific focus on optimising contrast agent concentration for high-resolution imaging in less than one night. This approach provides flexibility by allowing imaging of previously extracted specimens and maximising data acquisition from each sample, thereby reducing the number of animals required. Finally, we demonstrate that the proposed optimised protocol enables the acquisition of high-resolution dMRI data within a practical time frame (e.g., overnight), yielding reliable diffusion metrics, local orientation maps and whole-brain tractography of mouse white matter and allowing assessment of the influence of SNR on diffusion metrics and tractogram reconstruction.

2 | Methods

2.1 | Animal Preparation

A total of 20 female C57BL/6 mice (body weight = 25–39 g, age = 8–54 weeks, Charles River, France) were used. All experiments were approved by the local ethical committee (Animal Care and Use Institutional Ethics Committee of Bordeaux, #13152). Animals were anesthetised with isoflurane (Vetflurane, Virbac, France), and a solution of analgesics (100-mg ketamine/kg of body weight and 20-mg xylazine/kg) was administered via intraperitoneal injection. Anaesthesia depth was verified by absence of the pedal withdrawal reflex prior to transcardial perfusion. Perfusion was performed with phosphate-buffered 4% paraformaldehyde (PFA), and brains were fixed overnight at 4°C. After extraction, samples were stored in PBS 1× containing 0.01% sodium azide until imaging.

2.2 | MRI Hardware

All imaging experiments were performed using a horizontal 7-Tesla Bruker Biospec 70/20 system (Ettlingen, Germany) preclinical scanner with the Paravision 6.1 software. This system was equipped with a 12-cm gradient insert capable of 660 mT/m maximum strength and 110-µm rise time. A volume resonator was used (75.4-mm inner diameter, active length 70 mm) for excitation. A four-element (2 × 2) phased array surface coil (outer dimensions of one coil element: 26 × 16 mm^2, total outer dimensions: 26 × 21 mm^2) was used for the T_1, T_2 and diffusion-weighted data acquisition of Groups 1, 2 and 3. The diffusion-weighted data for the Group 4 was used with a surface cryoprobe coil four-element (2 × 2) with the Paravision 360 3.5 software. Each brain was placed in a sealed plastic Falcon tube filled with PBS, thereby preventing dehydration and minimise susceptibility artefacts. The tube was positioned with the rostrocaudal brain axis aligned parallel to the B0 field, centred within the RF coil and maintained at 25°C. Note that the cryoprobe was used for the acquisition of the final experimental group only, as this hardware became available after data collection for the earlier groups had already been completed.

2.3 | Experimental Design

The 20 mouse brains were divided into four groups involved in the different experiments to establish the optimal 3D SE-msEPI diffusion protocol (Figure 1).

Group 1 (*n* = 5) Brains were extracted using the perfusion and fixation protocol described above (§2.1, PBS followed by 4% PFA), without Gd-DOTA and were first imaged to determine the optimal number of shots to be used in the 3D SE-msEPI diffusion protocol. To this end, an air bubble was intentionally left near the brain to estimate the effect of the shot number on artefact dropout (see §2.5 for details). Subsequently, the same brains were immersed in 1 L of PBS containing increasing concentrations of Gd-DOTA (0–4 mM in 0.5-mM increments; Gd-DOTA, Dotarem, Guerbet, France), 1 week per step. After each incubation step, T_1 and T_2 maps were acquired.

Group 2 (*n* = 9) Brains were extracted using the same protocol as Group 1 and were incubated in 1 L PBS containing 3-mM Gd-DOTA for 14 days. T_1 and T_2 maps acquisition was performed just before the incubation with Gd-DOTA and after 1, 3, 7 and 14 days of incubation. Thereafter, brains were incubated in PBS without Gd-DOTA for 30 days, with imaging performed periodically (1, 3, 7, 14 and 30 days) to follow the contrast agent clearance. The PBS was changed after each acquisition.

Group 3 (*n* = 3) During euthanasia, a bolus of 15 mL PFA then a bolus of 15 mL PBS containing both 3-mM Gd-DOTA was transcardially injected. Brains were then extracted and stored overnight in 3-mM Gd-DOTA before T_1 and T_2 mapping.

Group 4 (*n* = 3) Brains were extracted using the same protocol as Groups 1 and 2 and were first imaged with the optimised 3D SE-msEPI diffusion protocol (12 shots), without Gd-DOTA, using two TR: 1500 ms (41 h 36-min acquisition) and 400 ms (11 h 06-min acquisition). Thereafter, the brains were incubated in 3-mM Gd-DOTA for 1 week and rescanned (with the TR = 400 ms acquisition). After another week of incubation in PBS without contrast agent, the brains were scanned again with the same protocol (TR = 400 ms). These four conditions will hereafter be referred to as 'long-TR baseline.' 'short-TR baseline,' 'Gd-DOTA enhanced' and 'postwashout,' respectively.

2.4 | MRI Acquisitions

For T_1 measurements (Groups 1, 2 and 3), a FAIR-RARE (Flow-sensitive Alternating Inversion Recovery) imaging sequence was employed with the following parameters: FOV = 20 × 20 mm; resolution = 156 × 313 μm^2; a slice of 1.5 mm thickness, first TI = 7 ms; 128 TI incremented by 10 ms; one average; TR = 5000 ms; TE = 2.86 ms; slice orientation = axial; reception bandwidth = 100,000 Hz; RARE factor = 8. The total scan time was ~1 h and 25 min.

For T_2 measurement (Groups 1, 2 and 3), a MESE (multiecho spin echo) imaging sequence was employed with the

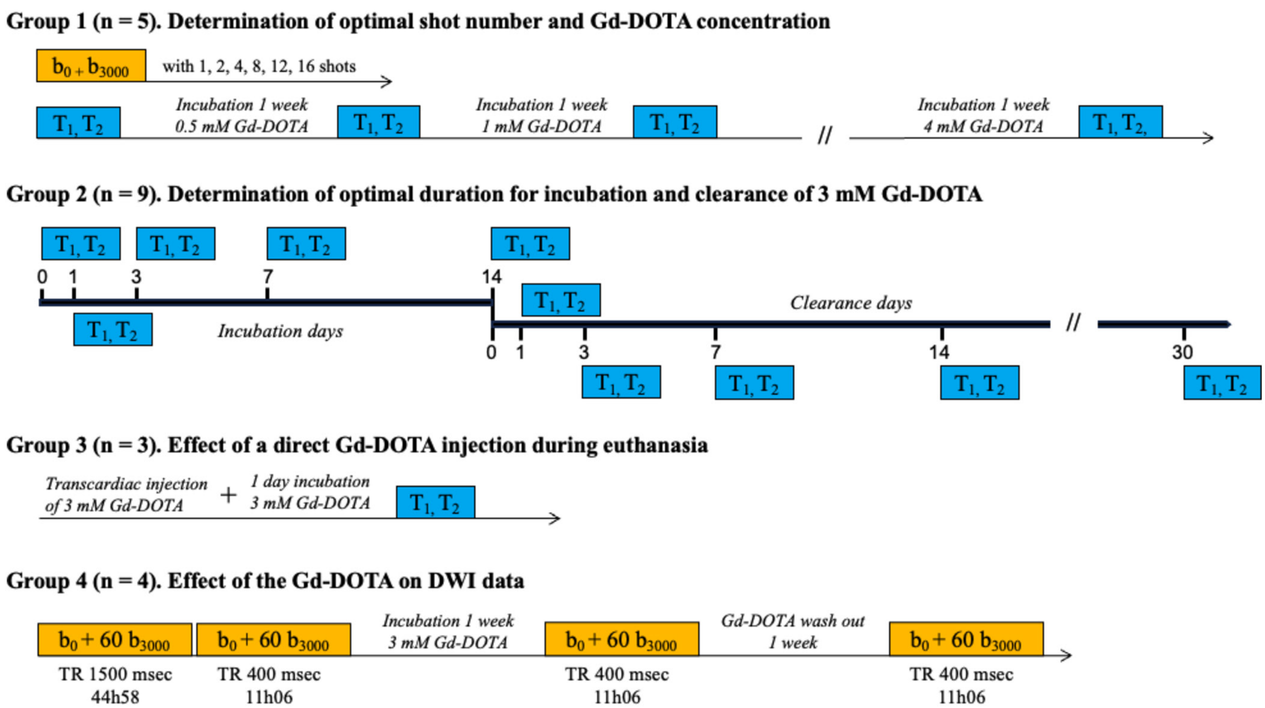

FIGURE 1 | Experimental design. The study includes 20 mouse brains divided into four groups. Depending on the experiments, b_0 and b_{3000} diffusion-weighted images (orange box) or T_1 and T_2 (blue box) images were acquired.

parameters: FOV = 20 × 20 mm; resolution = 156 × 156 µm; 1 slice of 1.5 mm thickness, TR = 5000 ms; inter-echo = 3.5 ms; one average; echo images = 128; slice orientation = axial; reception bandwidth = 150,000 Hz; flip angle = 90°. The total scan time was ~10 min.

Diffusion imaging (Groups 1 and 4) was performed using a 3D diffusion-weighted SE-msEPI sequence. The acquisition matrix was 192 × 128 × 128 over a 20 × 13 × 13-mm^3 field of view (FOV), resulting in a 104 × 102 × 102-µm voxel size. The multi-shot segmentation during the 3D SE-msEPI acquisition was applied along the phase-encoding dimension (128).

Formalin fixation affects tissue relaxation and also influences water diffusivity [18].

The diffusion-weighted gradients were designed with a high b value of 3000 s/mm^2 (b_{3000}), which is approximately three to four times higher than typically used in in vivo studies [11, 18, 19]. This choice was made to compensate for the reduced diffusion coefficient caused by tissue fixation in ex vivo samples [18]. To limit duty cycle and heat generation during the continuous 11-h acquisition, only 70% of the maximum gradient power was used. Consequently, the gradient duration (δ) was set to 5 ms and the diffusion time (Δ) to 10 ms. Furthermore, the vendor drift compensation option was applied to update the resonance frequency during acquisition, further mitigating intervolumes (3D volumes per direction) fluctuations.

In Group 1 (n = 5), the diffusion-weighted imaging (DWI) protocol included one nondiffusion weighted acquisition (b_0) and a single diffusion weighted direction at b_{3000}. Acquisition parameters were varied to determine the optimal protocol, including TR and the number of EPI shots, which dictated the TE. TE value was consistently minimised (Table 1).

The acquisition durations reported in Table 1 correspond to extrapolated total scan times, calculated as if a full diffusion protocol (5 b_0 images and 60 diffusion directions) had been acquired using the same acquisition parameters. This extrapolation was performed to allow direct comparison with the full multidirection DWI protocol used in Group 4.

In Group 4 (n = 3), the DWI acquisition began with five b_0 images, followed by 60 diffusion directions at b_{3000}. The number of EPI shots was set to 12 with TE = 20.4 ms, and two TR values were used: 1500 and 400 ms, resulting in acquisition durations of 41 h 36 min and 11 h 06 min, respectively. In the short-TR baseline and postwashout condition, the flip angle was adjusted to 45° according to the Ernst equation in order to optimise the signal, whereas 90° was used for long-TR baseline and Gd-DOTA–enhanced conditions.

2.5 | MRI Data Processing

As mentioned in §2.3, for Group 1 (n = 5), an air bubble was intentionally left near each brain in order to create an artefact on the EPI images. A region of interest (ROI) around this air bubble artefact was manually drawn on the EPI image acquired with one shot for each brain. The average signal within this ROI was calculated on each image acquired with 1, 2, 4, 8, 12 and 16 shots. For normalisation purposes, the average signal was divided by the average signal of the acquisition with 16 shots for each brain.

For each brain, a ROI was manually drawn in the isocortex on the single b_{3000} image and its anatomical correspondence was verified across shots. In addition, a second ROI was placed in a background outside the brain. The SNR was computed between the mean signal within the isocortex ROI and the standard deviation of the background noise ROI. The isocortex region was chosen because there is no preferential diffusion, so it will be minimally affected regardless of the direction.

In a second time, the same brains were immersed in 1 L PBS containing increasing concentrations of Gd-DOTA (0–4 mM in 0.5 mM increments, 1 week each time). Voxel-wise T_1 and T_2 relaxation time constant maps were first computed by fitting the signal evolution according to TI or TE, respectively. Mean relaxation time constants were then extracted by averaging voxel values within manually defined ROIs placed in the isocortex (cortical grey matter), corpus callosum (white matter) and thalamus (deep grey matter) using a custom-built processing software based on Igor Pro64, Wavemetrics (Version 8.04). Care was taken to maintain similar ROI positions across brains to ensure proper comparison of values. For these experiments, T_1 and T_2 values, as well as mean relaxation time constants and corresponding standard deviations (SD) across brains, were calculated for the three ROIs.

The same approach as for Group 1 was applied to delineate the three ROIs (isocortex, corpus callosum and thalamus) and to extract relaxation time constants from the T_1 and T_2 maps acquired in the brains of Group 2 (n = 9) and Group 3 (n = 3).

The DWI data acquired in Group 4 (n = 3) benefited from the development of MouseFlow, a pipeline designed specifically to automate the preprocessing and analysis of dMRI data for mouse brain imaging [20]. MouseFlow is adapted from the TractoFlow pipeline initially designed for the human brain [21]. Every default parameter was meticulously adjusted for optimal performance on ex vivo mouse brain data. It involves using BrkRaw (https://github.com/BrkRaw/brkraw), FSL [22], MRtrix3 [23], ANTs [24] and DIPY-based SCILPY [25, 26] (http://github.com/scilus/scilpy) tools. In brief, MouseFlow pipeline first integrates denoising, correction for eddy currents, brain mask extraction and N4 bias correction steps. Subsequently, the processed DWI data are utilised to calculate

TABLE 1 | Parameters of the 3D diffusion-weighted SE-msEPI sequence for full diffusion protocol (5 b_0 images and 60 diffusion directions).

TR = 1500 ms	1 shot	2 shots	4 shots	8 shots	12 shots	16 shots
TE (ms)	75.6	44.4	30.0	22.3	20.4	20.1
Acquisition	3h28	6h56	13h52	27h44	41h36	55h28

diffusion tensor imaging (DTI) metrics [27–29]. Note that fractional anisotropy (FA) was the only DTI-derived metric computed and analysed in the present study. The extraction of FA values within specific anatomical ROIs is conducted within the Allen Mouse Brain Atlas [30] (AMBA). The volumetric average AMBA template at 100-μm isotropic voxel resolution was registered to each mouse's DWI native space using ANTs. Both the affine transformation and nonlinear deformations were applied to warp the AMBA parcellation to each DWI native space.

For the median FA measurement, the ROIs used corresponded to those for T_1 and T_2 mapping. However, in accordance with the AMBA atlas, the corpus callosum was grouped into a single large region, as was the isocortex. In addition, the thalamus was considered as a whole.

The residual map R was defined voxel-wise as the difference between the measured diffusion-weighted signal and the signal predicted by the fitted DTI model. Specifically, for each voxel, a diffusion tensor was estimated from the diffusion-weighted measurements, and the expected signal attenuation for each diffusion-encoding direction was computed using the standard mono-exponential DTI signal model. The residual corresponds to the absolute difference between the measured and model-predicted signals, averaged across diffusion directions. This residual provides an indicator of how accurately the DTI model characterises the observed signal [31]. For the residual map, only white matter ROIs were selected: the internal capsule (int), the anterior commissure (aco) and the body of the corpus callosum (ccb). Those average measurements were performed in the subject space; then, the images were moved to a common space to allow for better visual comparison.

For each of the three conditions (short-TR baseline, Gd-DOTA enhanced and postwashout), the FA map was registered within the AMBA space. The difference from the reference image (long-TR baseline) was then calculated voxel wise (FA_subject − FA_ref divided by FA_ref), and the result was multiplied by 100 to express the relative variations in percentage units.

White matter pathways of the mouse brains were reconstructed using diffusion MRI tractography. Whole-brain tractograms were generated through the MouseFlow pipeline using a streamline-based high-angular-resolution diffusion imaging (HARDI) algorithm optimised for angular precision [32, 33]. The tracking is following the local orientation distribution function (ODF), with direction chosen deterministically as peaks of the spherical harmonic Q-ball model [34]. This approach ensures robust tracking in regions of complex fibre architecture while avoiding probabilistic dispersion. Parameters were specifically tuned for the mouse brain: an angular threshold of 45°, a step size equal to one-tenth of the isotropic voxel size (i.e., 10 μm here), a seeding in a WM mask derived from AMBA white matter labels (1 seed per voxel) and a tracking in the whole-brain mask.

The effect of the current optimised protocol was first assessed globally by calculating the seeding efficiency defined as the ratio of the number of streamlines obtained in the whole-brain tractogram to the number of seeded voxels, across the four conditions used in Group 4 ('long-TR baseline,' 'short-TR baseline,' 'Gd-DOTA enhanced' and 'postwashout'). The seeding efficiency parameter was chosen as a quantitative metric to assess tractography performance across several experimental conditions. Specifically, it reflects the proportion of initiated streamlines that successfully propagate through the diffusion field under a given acquisition. Higher seeding efficiency indicates more stable and coherent local diffusion orientations. Seeding efficiency thus provides a useful indicator of tractography robustness and sensitivity to image quality (e.g., SNR), which was the focus of this study.

The same approach was applied with a specific WM bundle, namely, the anterior commissure (AC), by estimating the seeding efficiency as the ratio of the number of AC streamlines extracted from the whole-brain tractogram to the mid-sagittal AC volume in each mouse's DWI native space. Extraction of the AC is based on the AMBA olfactory (aco) and temporal (act) AC limbs. They were combined into a single ROI representing the full AC, from which a mid-sagittal AC inclusion ROI was derived by keeping only a thin slab (5 voxels) centred on the brain midline in the X (left–right) axis. The goal was to ensure that only fibres crossing at the midline segment of the AC were retained. An exclusion ROI was defined from AMBA structures surrounding the AC, namely, the median preoptic nucleus (MEPO), medial preoptic area (MPO), paraventricular nucleus of the thalamus (PVT), septofimbrial nucleus (SF) and the corpus callosum composed of the body (ccb), genu (ccg), rostrum (ccr), splenium (ccs), anterior (ec, fa) and posterior (ee, fp) forceps AMBA labels.

3 | Results

3.1 | Optimal Number of Shots

As expected, the size of the bubble artefact (surrounded in yellow in Figure 2A) visually decreases as the number of shots increases. This translates into an increase of the signal in the ROIs (Figure 2B). The values increase from 0.65 ± 0.09 for 1 shot to 0.94 ± 0.03 for eight shots, then to 0.97 ± 0.007 for 12 shots. The values for 12 shots are very close to the reference image, that is, 16 shots. Although the artefact is still visible on diffusion images (Figure 2C), it becomes negligible after 12 shots. These results demonstrate that the use of 12 shots or more allows for a substantial reduction in artefacts, consistent with medium to large effect sizes, justifying their use for improved segmentation accuracy and reduced sensitivity to artefacts.

In addition to reducing artefacts in areas with air bubbles, increasing the number of EPI shots also improved the SNR thanks to a shorter TE. The average SNR on the diffusion-weighted image at one shot was 6.1 ± 1.5, while after 12 shots it increased to 17.5 ± 1.4. At 12 shots, the signal reached a level sufficient for reliable imaging, with a marginal improvement at 16 shots (signal-to-noise ratio $= 18 \pm 3.4$). These results demonstrate that 12 shots (and therefore a TE $= 20.4$ ms) offer a significant gain in signal-to-noise ratio, with little additional benefit from more shots.

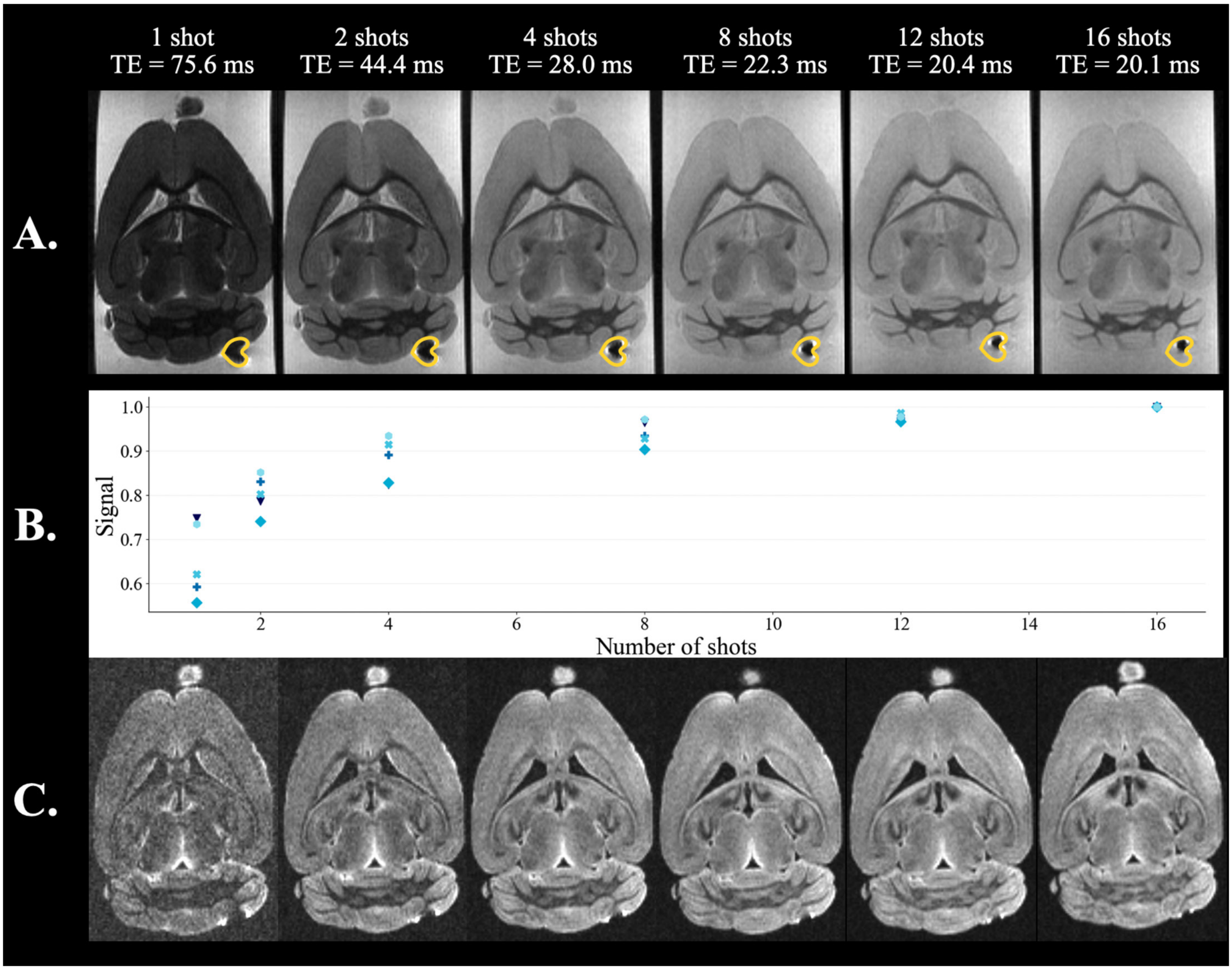

FIGURE 2 | Evaluation of the influence of the number of shots on diffusion-weighted spin-echo EPI images acquired at b_0 (A) and b_{3000} (C) in one brain of Group 1. The ROI where the mean signal of the bubble was calculated is delineated in yellow. B represents the artefact signal value on b_0 images that increases in the ROI as the number of shots increases. $n = 5$ for each measurement.

3.2 | Optimal Usage of Gd-DOTA

Figure 3A shows the linear fits between mean relaxation rates ($1/T_1$ and $1/T_2$) and Gd-DOTA concentration, computed in the isocortex ROI across nine brains. These values were used to estimate the relaxivities r_1 and r_2. The coefficients of determination (R^2) were close to unity (0.97 and 0.96 for the T_1 and T_2, respectively) indicating a strong linear relationship between Gd-DOTA concentration and the measured relaxation rates ($1/T_1$ and $1/T_2$), even when the Gd-DOTA was incubated after extraction. The relaxivities in the isocortex were estimated at $r_1 = 2.66\,\text{mM}^{-1}.\text{s}^{-1}$ and $r_2 = 4.54\,\text{mM}^{-1}.\text{s}^{-1}$. Using these data, the simulation tool described in Barrett et al. (2023) [35] predicted an optimal incubation concentration of Gd-DOTA of 2.8 mM for the selected parameters (TE = 23.4 ms, TR = 400 ms), as indicated by the circle in Figure 3B. The tool optimises the SNR of a spin-echo sequence as a function of TE, TR and Gd-DOTA concentration. T_1 and T_2 relaxation time constants are modelled as functions of Gd-DOTA concentration (Figure 3A), and the SNR is calculated using an analytical expression that accounts for T_1 recovery, T_2 decay and normalisation by acquisition time. For each TE-TR combination, the Gd-DOTA concentration that maximises the SNR is determined.

For practical purposes, a concentration of 3 mM was selected (for which representative T_1, T_2 maps obtained for one of the five Group 1 mouse brains are displayed) (Figure 3C).

After establishing the optimal concentration of Gd-DOTA to incubate (i.e., 3 mM), the minimum incubation time required to achieve homogeneous diffusion of the contrast agent across the different brain tissues was determined. Thereafter, the minimum wash-out time necessary to fully eliminate the Gd-DOTA from the brain was assessed, thereby enabling its subsequent use in additional protocols (histology, immunohistochemistry and light-sheet imaging) (Figure 4).

After the first 24 h of incubation, Gd-DOTA induced a pronounced decrease in T_1 values in the three regions: from 1283 ± 77 to 131 ± 13 ms in the isocortex, from 1153 ± 93 to

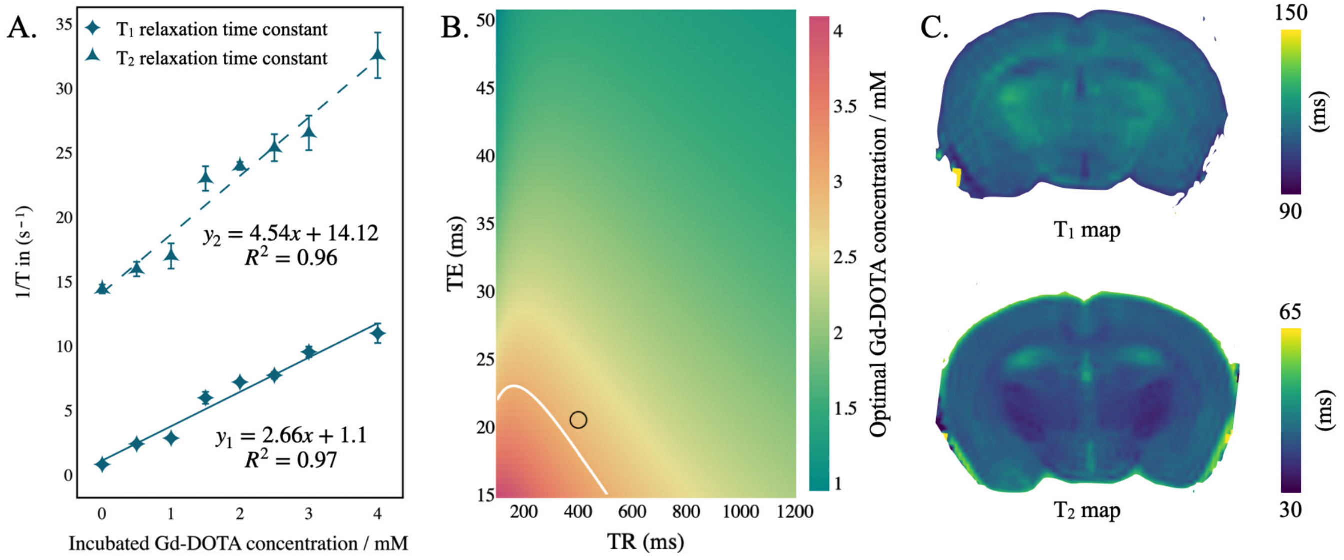

FIGURE 3 | (A) r_1 and r_2 MRI relaxivity plots obtained from a brain incubated with increasing concentrations of Gd-DOTA. The diamonds, triangles and circles correspond to the mean relaxation time constants T_1 and T_2, respectively, obtained on the isocortex of nine brains (Group 1). Linear regressions are shown. (B) Plot representing the concentration of Gd-DOTA that provides the maximum SNR efficiency for each combination of TE and TR as computed in [35]. The circle shows the optimal concentration of Gd-DOTA to use when a TE/TR = 20.4/400 ms are set in the msEPI sequence. (C) Representative relaxation maps showing the spatial distribution of T_1 and T_2 and relaxation time constants for the same brain incubated with 3-mM Gd-DOTA during 1 week.

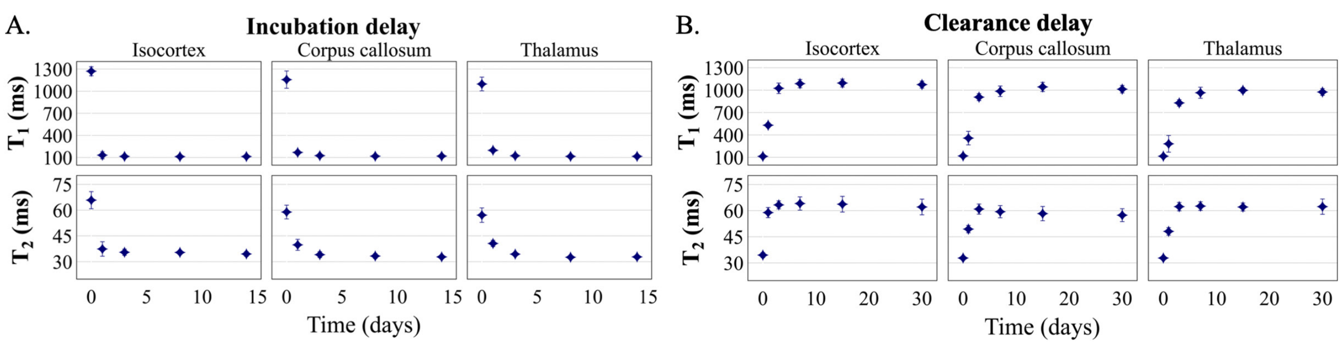

FIGURE 4 | Evolution of T_1 and T_2 relaxation time constants in the isocortex, corpus callosum and thalamus during incubation with 3-mM Gd-DOTA (A) and during clearance in PBS (B). Data points and error bars show the mean and the standard deviation in the region of interest for nine brains (Group 2).

169 ± 20 ms in the CC and from 1104 ± 51 to 199 ± 30 ms in the thalamus. Beyond 7 days, relaxation time constants stabilised (Figure 4A). Overall, T_1 values strongly decreased by 82%–89% across the regions of interest. Similarly, T_2 values decreased by 28%–43% after 24 h of incubation. These results indicate that optimal MR imaging conditions are achieved at least after 1 week of incubation with 3-mM Gd-DOTA.

Both relaxation time constants increased progressively after transferring the doped brains into a large volume of PBS, reflecting the clearance of the Gd-DOTA from the tissue (Figure 4B). A sharp recovery was observed during the first days, after which values plateaued across all regions by approximately day 15. After 30 days, T_1 and T_2 values approached those measured before Gd-DOTA exposure ($t = 0$ in Figure 4A). On average, 87.3%–89.7% of the baseline T_1 and 98.5%–105.7% of the baseline T_2 were restored after 30 days of wash-out. Across all brains, the T_1 and T_2 values for each ROI converged to the same level after one week of incubation.

To evaluate whether passive incubation could achieve a similar level of penetration within the brain tissue as direct delivery of the contrast agent, passive incubation was compared with vascular perfusion of 3-mM Gd-DOTA during euthanasia (Group 1 vs. Group 3).

In the isocortex, perfused brains ($n = 3$) showed $T_1 = 112 \pm 3$ ms and $T_2 = 34 \pm 1$ ms, while incubated brains ($n = 5$) yielded $T_1 = 111 \pm 2$ ms and $T_2 = 34 \pm 1$ ms. In the corpus callosum, perfused brains exhibited $T_1 = 112 \pm 3$ ms and $T_2 = 32 \pm 2$ ms, compared with $T_1 = 117 \pm 4$ ms and $T_2 = 35 \pm 1$ ms in incubated brains. In the thalamus, perfused brains showed $T_1 = 115 \pm 4$ ms and $T_2 = 32 \pm 1$ ms, whereas incubated brains had $T_1 = 115 \pm 3$ ms and $T_2 = 32 \pm 1$ ms. Overall, these results indicate that T_1 and T_2 values are highly comparable between perfused and incubated brains across all ROIs, suggesting that either method of Gd-DOTA administration can be used effectively, depending on experimental constraints.

3.3 | Effects of Gd-DOTA on Diffusion Images and Tractography

The FA images acquired with a short TR and after washout showed reduced SNR, resulting in decreased image quality (Figures 5A). This noise decreases with long TR and with Gd-DOTA. Gd-DOTA not only provides a better SNR but also brings the SNR values closer to those of the reference condition, namely, long-TR baseline.

As expected, for a given TR Gd-DOTA increases the SNR and thus approaches the SNR of the long-TR baseline with an average value of 153.4 ± 6.87, 155.5 ± 19.44 and 100.2 ± 9.86 compared to 205.3 ± 19.95, 161 ± 9.36 and 116.7 ± 8.61 for the three ROIs. For short-TR baseline and postwashout, the values are lower (36.8 ± 5.42, 29.2 ± 3.28, 22.3 ± 2.74 versus 42.2 ± 15.52, 33.4 ± 15.63, 25.1 ± 10.68, respectively).

Regardless of the region, the long-TR baseline and Gd-DOTA–enhanced protocols yielded higher SNR values, reflecting increased longitudinal magnetisation recovery (Figure 5B). FA estimates using enhanced Gd-DOTA are similar to those using long TR. At the same time, a decrease in FA was observed after washing (0.17 ± 0.018 and 0.12 ± 0.009), corresponding to a decrease in SNR. It should be noted that, regardless of the condition, the standard deviation appears to be identical.

In order to go beyond a regional analysis, voxel-wise normalisation of the differences was applied (Figure 5C). Under low SNR conditions (short-TR baseline and postwashout), a negative bias is observed, shifting the distributions to the left and indicating overall lower FA values than the reference. Conversely, the Gd-DOTA–enhanced condition is more closely centred around zero, indicating a smaller difference from the reference (long-TR baseline).

The residual maps of long-TR baseline and Gd-DOTA enhanced showed a more homogeneous image and smaller R than short-TR baseline and postwashout, reflecting better image quality (Figure 6A). This is reflected in the white matter regions, where R is 16.5 ± 2.1 in the long-TR baseline and 11.5 ± 1.75 in Gd-DOTA enhanced, whereas it increased to 28 ± 8.02 in the short-TR baseline and 18 ± 5.18 in post washout (Figure 6B).

The standard deviation (SD) of the residuals is correlated with SNR: The SD is larger for the protocols with lower SNR.

This protocol confirmed that low SNR acquisitions result in underestimated FA values and increased residual variance.

Finally, the effects of the optimised protocol on the quality of WM fibre reconstruction were examined using tractography. Seeding efficiency values were compared across the four acquisitions (long-TR baseline, short-TR baseline, Gd-DOTA enhanced and postwashout) using whole-brain tractograms from Group 4. Seeding efficiency was consistently lower under low-SNR conditions (short-TR baseline and postwashout) compared to high-SNR conditions (long-TR baseline and Gd-DOTA enhanced) (Figure 7A). Taken together, these results indicate that tractography generates a comparable number of streamlines between the long-TR baseline and Gd-DOTA–enhanced conditions and consistently fewer streamlines in low-SNR conditions.

Seeding efficiency measured within the AC did not show an overall effect across conditions. Nevertheless, qualitatively, inspection of Figure 7B in a representative mouse brain shows that both whole-brain tractograms and the streamlines constituting the AC appear more regular, continuous and anatomically consistent with the AC anatomy [36] for the long-TR baseline and Gd-DOTA–enhanced conditions. In contrast, tractograms obtained under short-TR baseline and postwashout conditions exhibit more irregular and fragmented streamlines, deviating from the expected AC anatomy.

4 | Discussion

Developing a fast and robust diffusion sequence for the ex vivo mouse brain is challenging due to the small brain size, which requires high spatial resolution and, a high signal-to-noise ratio. In this work, we established an optimised protocol combining a Spin-Echo multishot EPI (SE-msEPI) diffusion sequence with an optimal concentration of Gd-DOTA to achieve this balance. The complete protocol is provided in Supplementary Figure S1.

4.1 | Diffusion MRI Sequence Optimisation

When combined with Gd-DOTA, the 3D SE-msEPI sequence reduced susceptibility-related artefacts and shortened the acquisition time by a factor of four compared with a 3D SE single-shot EPI sequence. Achieving this performance required careful optimisation of both the EPI segmentation strategy and the contrast agent concentration.

Although SE-msEPI intrinsically involves longer scan times than SE-ssEPI, this limitation is less critical for ex vivo imaging, where scan duration is not constrained by physiological considerations. Segmenting k-space over multiple excitations reduces echo time and mitigates distortions caused by magnetic field inhomogeneities, which are particularly pronounced in ex vivo samples due to air bubbles and tissue–air interfaces. Increasing the number of shots further improves robustness to susceptibility effects by shortening TE. While residual distortions may persist in regions with large susceptibility gradients, these effects are substantially reduced compared with single-shot acquisitions [37, 38].

Acquisition time can be further shortened by reducing the TR; however, this introduces a trade-off between signal recovery that depends on tissue T_1 relaxation time and targeted SNR. Increasing the Gd-DOTA concentration shortens T_1 and enables shorter TR, but excessive concentrations also reduce T_2 and may lead to signal loss. At a fixed TE of 20.4 ms, a TR of 400 ms combined with a Gd-DOTA concentration of 3 mM provided a favourable compromise between SNR, relaxation effects and acquisition time, enabling routine overnight scans while preserving image quality.

With this configuration, the total acquisition time was reduced to approximately 11 h for 100-μm isotropic resolution with 60

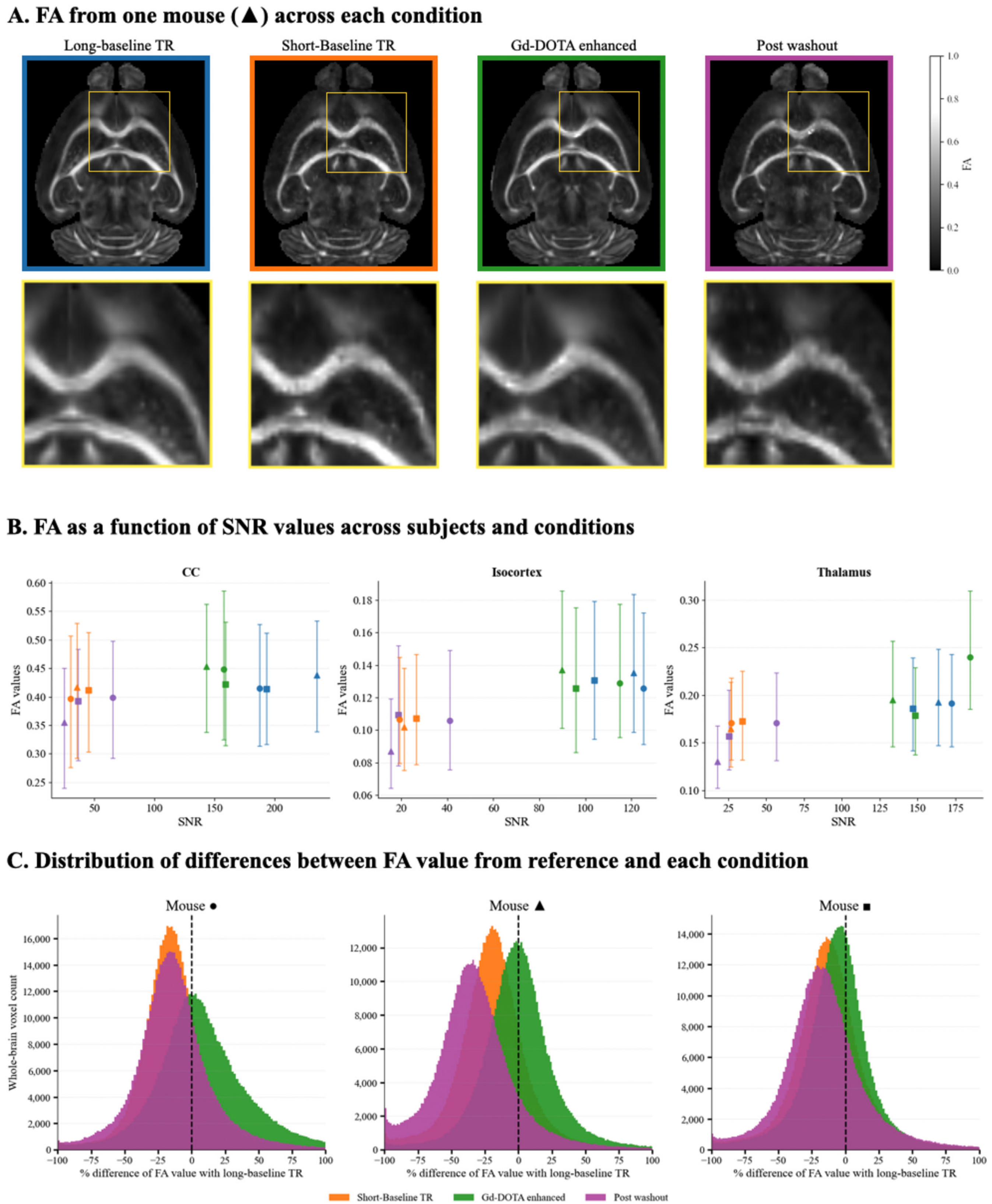

FIGURE 5 | Evaluation on FA values. (A) Top: FA (axial slice) from the subject 3 (triangle) for the four acquisition conditions in the common space. Bottom: zoom-in for the square region. (B) Effect of SNR on median FA values measured in three regions of the three subjects: corpus callosum (CC), isocortex and thalamus. (C) FA difference for all voxels with the long-TR baseline for each subject.

diffusion directions, representing a substantial improvement over previous studies requiring approximately 40 h [13, 14]. This reduction is particularly advantageous for studies involving large sample cohorts. The optimised protocol provided sufficient SNR for robust estimation of diffusion metrics such as FA and enabled the generation of anatomically plausible tractograms

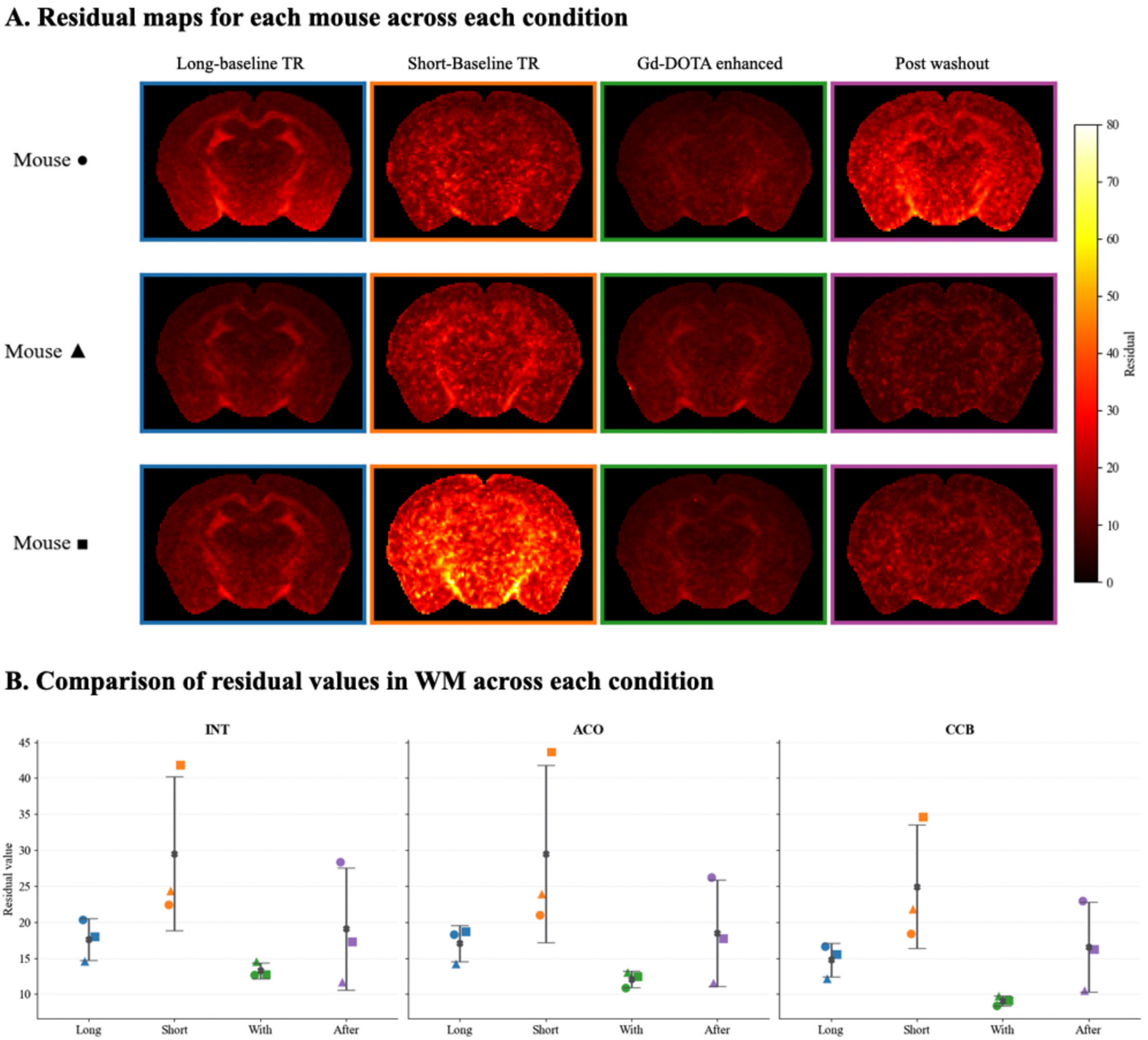

FIGURE 6 | Evaluation of residual values. (A) Residual maps (coronal slice) from all mice under the four conditions in the common space. (B) Residual values measured in three white matter regions: internal capsule (INT), anterior commissure (ACO) and corpus callosum body (CCB). Each subject is represented by a different pictogram, whereas each condition is represented by a colour.

comparable in quality to those obtained with considerably longer acquisitions.

Alternative acceleration strategies, such as compressed sensing, have been used to compensate for the long acquisition times associated with SE diffusion-weighted imaging [6, 12, 39]. The segmented EPI approach adopted here additionally benefits from fast, noniterative reconstruction, resulting in short reconstruction times. Nevertheless, potential EPI-related blurring should be considered [40]. At identical nominal resolution (100 μm), no measurable degradation of effective spatial resolution was observed in the contrast-enhanced condition compared with the long-TR reference acquisition, despite the T_2* shortening induced by the contrast agent. At higher spatial resolutions, further optimisation of the number of shots or Gd-DOTA concentration may be required, at the expense of increased acquisition time.

Susceptibility-induced geometric distortions are a known limitation of diffusion-weighted MRI and are generally addressed using approaches such as TOPUP, which relies on complementary acquisitions with inverted phase encoding [37]. In the present study, such correction was not applied systematically. This choice was motivated by the ex vivo experimental conditions, in which distortions related to B0 inhomogeneity are generally greatly reduced thanks to optimised shimming, sample immersion and the absence of air-tissue interfaces [38, 39]. Nevertheless, residual distortions may persist, particularly in the presence of bubble-related artefacts, although their visual impact decreases with increasing number of acquisitions, partly due to the reduction in background signal. The absence of phase-encoded inverted data therefore represents a limitation of the present work. Future studies could benefit from the systematic acquisition of these data to enable TOPUP correction,

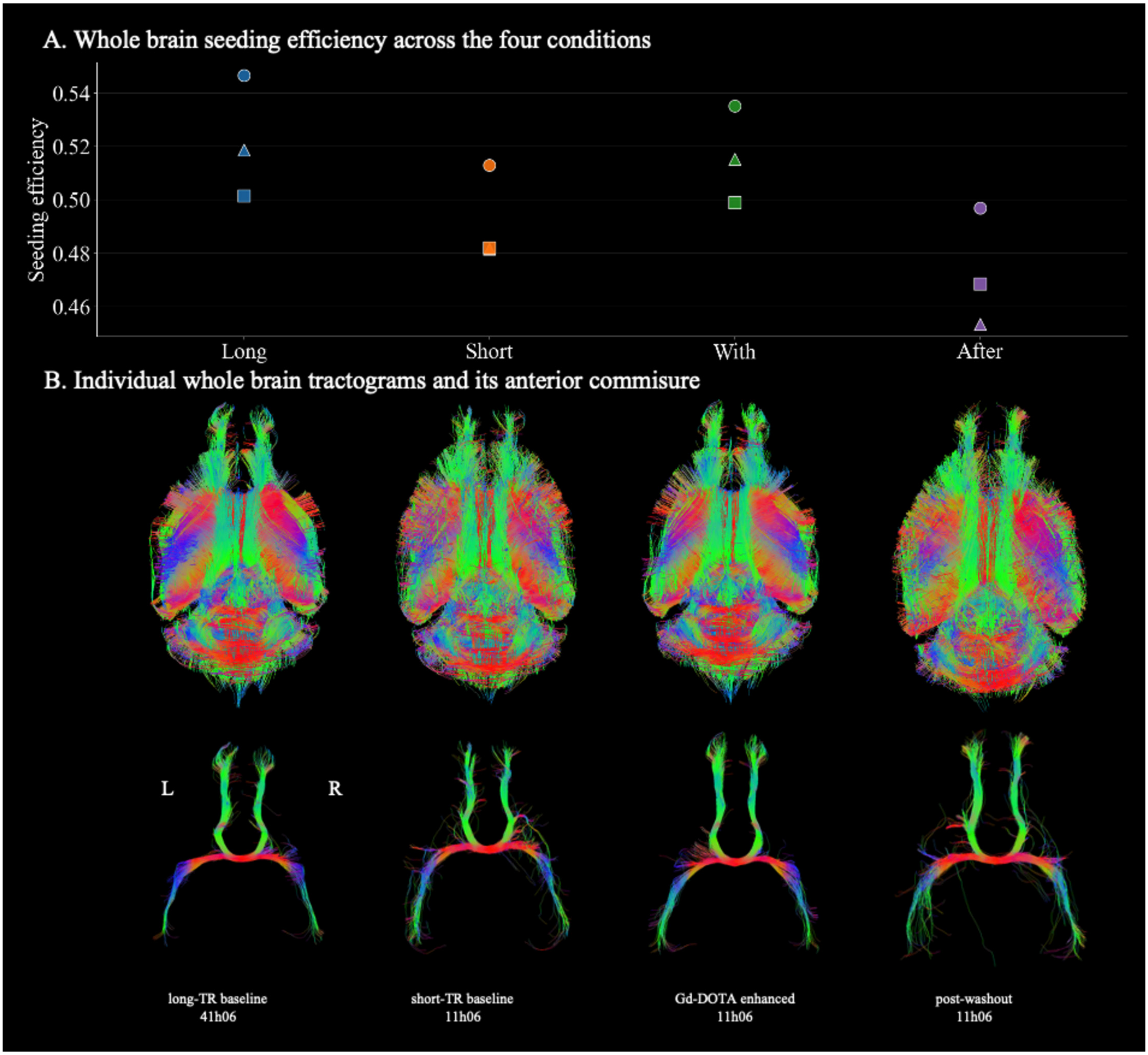

FIGURE 7 | (A) Seeding efficiency values were compared across the four conditions in three whole-brain tractograms (Group 4). (B) Tractography reconstructions under the four acquisition conditions in a representative mouse brain from Group 4. Top row: superior views of whole-brain tractograms. Bottom row: superior views of the anterior commissure (AC) extracted from the corresponding whole-brain tractograms. Reconstructions in the long-TR baseline and Gd-DOTA–enhanced conditions appear more continuous and anatomically faithful, whereas those in the short-TR baseline and postwashout conditions show fragmented and irregular streamlines.

which could further mitigate residual B0-related distortions and improve geometric fidelity, particularly in challenging regions.

4.2 | Gadolinium Concentration Optimisation

We determined that a concentration of 3-mM Gd-DOTA was optimal for ex vivo mouse brain diffusion imaging lasting one night. One week of incubation was necessary to reach a homogeneous diffusion throughout the tissue. This incubation time, mainly due to the large incubation volume relative to brain size, is an advantage over a previous study where the brain was incubated with the Gd-DOTA solution for 60 days [40]. Importantly, Gd-DOTA could be efficiently washed out within 2 weeks, allowing subsequent reuse of the same samples for complementary imaging protocols.

Perfusion of Gd-DOTA at euthanasia offers a rapid method to achieve uniform contrast within 24 h, but immersion incubation provides distinct advantages. It can be applied to brains that have already been extracted and avoids variability introduced by blood circulation. Thus, both approaches are valid, but immersion incubation offers greater flexibility for sample reuse.

In vivo, Mn^{2+} enters neurons through calcium [41, 42] channels or synaptic activity [43] ex vivo study [15]. Other contrast agents, such as MnCl, have been explored for MRI enhancement, but their mechanisms of tissue distribution differ from that of Gd-DOTA., whereas Gd-DOTA is a gadolinium chelate that does not rely on cellular uptake mechanisms. As a result, MnCl-based contrast enhancement is not readily compatible with the passive, diffusion-based incubation protocol used in the present.

4.3 | Effects of Gd-DOTA on Diffusion MR Parameters

We compared the optimised Gd-DOTA–enhanced acquisition (~11 h, TR = 400 ms) with the reference long-TR acquisition (~44 h, TR = 1500 ms). Gd-DOTA provided SNR levels comparable to the long-TR condition in all regions.

Formalin fixation is known to substantially reduce absolute water diffusivities compared with in vivo tissue, while generally preserving diffusion anisotropy and FA, as fibre organisation remains largely intact [18, 44, 45]. These effects arise from tissue cross-linking and reduced molecular mobility. In contrast, gadolinium-based contrast agents such as Gd-DOTA do not alter tissue microstructure but primarily affect MR relaxation properties, thereby improving SNR. Consequently, any influence of Gd-DOTA on diffusion metrics is expected to be indirect and mediated by SNR rather than by changes in diffusivity or anisotropy. Consistent with this interpretation, FA values obtained under Gd-DOTA–enhanced conditions closely matched those of the long-TR reference acquisition in our data.

Our data confirm the strong dependence of FA estimation on SNR. In the present study, low SNR was associated with a systematic underestimation of FA across both high- and low-FA regions, consistent with prior reports [46, 47]. However, it differs from earlier reports describing FA overestimation at moderate noise levels due to eigenvalue repulsion under conditions of constant diffusion tensor and longitudinal magnetisation [48, 49]. In contrast, the reduced FA observed here arises from short-TR acquisitions performed without Gd-DOTA, which lead to incomplete recovery of longitudinal magnetisation and a substantial reduction in SNR. Under these conditions, instability in principal eigenvector estimation results in increased orientational dispersion and an effective underestimation of FA, an effect that dominates over noise-induced FA overestimation. In the same vein, the study by Jones and Basser (2004) [50] showed that at high b values ($>1000\,s/mm^2$), a low SNR can lead to an underestimation of FA.

A recent study reported low-SNR-related FA biases limited to specific grey matter regions [51] and addressed this issue by increasing SNR through repeated acquisitions, with the associated challenges of image registration. In contrast, in our study the administration of Gd-DOTA shortened T_1, restored longitudinal magnetisation at short TR, and increased SNR, thereby improving tensor estimation. Consequently, FA values increased and approached those obtained under reference long-TR conditions. Overall, these findings indicate that the observed relationship between SNR and FA reflects a transition between different SNR regimes rather than a contradiction with the existing literature.

However, a suboptimal choice of the excitation flip angle for the short baseline TR and the postwashout conditions was made. A flip angle of 45° was selected, whereas 90° was used for the two other conditions. The SNR of the short-TR baseline and the postwashout conditions was therefore 82% lower than the long-TR baseline condition, whereas it could have been only 62% lower with a 90° excitation flip angle and only 45% lower with the supplement of the Ernst angle (135°) [52] (See supplementary Figure S2).

4.4 | Effects of Gd-DOTA on Tractography

The effects of the optimised acquisition protocol on WM tractography were evaluated in terms of both anatomical reliability and the quantitative accuracy of the diffusion information underlying streamline reconstruction. Anatomical reliability was assessed by combining tractography performance metrics with qualitative inspection of reconstructed white matter pathways.

Whole-brain tractograms generated with the optimised protocol (12 shots, TR = 400 ms, 3-mM Gd-DOTA) exhibited seeding efficiency values comparable to those obtained with the long-TR reference acquisition and consistently higher than those observed under low-SNR conditions (short-TR baseline and postwashout). This indicates that the optimised protocol provides sufficiently stable and coherent local fibre orientation estimates to support robust streamline propagation across the brain. In contrast, reduced seeding efficiency under low-SNR conditions reflects increased instability in local orientation estimation, limiting successful tract propagation.

Beyond global tractography metrics, qualitative inspection of reconstructed bundles further supported anatomical reliability. In particular, the anterior commissure, an anatomically well-characterised midline-crossing white matter tract, displayed continuous, symmetric and spatially coherent trajectories under both the long-TR baseline and Gd-DOTA–enhanced conditions (Figure 7B). These reconstructions are consistent with established descriptions of anterior commissural anatomy in the mouse brain [36]. Conversely, tractograms obtained under low-SNR conditions showed fragmented, irregular and discontinuous streamlines, suggesting artefactual tracking driven by noise rather than true anatomical features.

Quantitative accuracy of tractography was not assessed against a histological ground truth but was evaluated indirectly through the accuracy of the diffusion metrics that underpin streamline reconstruction. FA values obtained under the Gd-DOTA–enhanced condition closely matched those measured in the long-TR reference acquisition across multiple anatomically defined regions, including the corpus callosum, cortex and thalamus. Voxel-wise FA difference maps relative to the reference condition showed minimal bias under the optimised protocol, whereas systematic FA underestimation was observed under low-SNR conditions (Figure 5). In addition, residual maps revealed reduced discrepancies between measured and modelled diffusion signals in major white matter regions, indicating improved tensor fitting when using the optimised acquisition (Figure 6).

Taken together, we demonstrate that the optimised protocol enables the generation of tractograms that are anatomically plausible and based on quantitatively accurate diffusion tensor estimates. These findings highlight the value of the protocol for connectomics applications, where accurate reconstruction of long-range connections is critical. By providing reproducible and anatomically secured tractograms, the method improves both efficiency and biological interpretability in preclinical mouse models [53].

4.5 | Limitations and Future Directions

While immersion incubation enables reuse of samples, the 1-week incubation time may reduce throughput and could vary across fixation methods, brain sizes or disease models. Perfusion at euthanasia offers a faster alternative but lacks the flexibility of retrospective sample preparation.

The first limitation is the modest sample size, which reduces statistical power. Larger cohorts would strengthen validation. Second, the choice of $b = 3000\,\mathrm{s/mm^2}$, based on prior literature [13], represents a compromise between sensitivity to micro-movements of water molecules while having enough SNR and being able to generate accurate tractograms [54]; additional studies could refine this choice to get a higher precision in tractograms. Finally, optimisation was performed at 7T using specific hardware. Adaptations might be necessary for other field strengths, coil designs and experimental contexts.

Another limitation of the present optimised protocol is the presence of Gd-DOTA, which may affect subsequent histological analysis on the scanned mouse brains. The elimination phase was monitored quantitatively using T_1 and T_2 measurements, which progressively returned to baseline values, indicating clearance of the contrast agent from the tissue. However, the potential impact of residual Gd-DOTA on subsequent histological analyses was not directly evaluated in this study and is therefore not demonstrated here. To our knowledge, there is no evidence in the literature indicating incompatibility between Gd-DOTA and standard histological procedures. Nevertheless, as this aspect was not systematically assessed, it should be considered a limitation of the study and an important avenue for future research.

A limitation of this study is that all experiments were performed on healthy mouse brains. Consequently, the sensitivity of the proposed protocol to disease-related alterations of specific white matter tracts was not assessed. Future work will be required to evaluate its performance in pathological models involving focal or subtle white matter changes. In addition, while the data quality achieved here is sufficient for tensor-based diffusion metrics and tractography, the suitability of the protocol for advanced microstructural diffusion models (NODDI [55] or DKI [56]) was not evaluated and remains an important direction for future investigation.

Further investigations include exploring faster acquisition methods (like undersampling the k-space) combined with advanced reconstruction methods such as compressed sensing [6, 12, 57] or low rank [58, 59] and integrating diffusion MRI data with histology or light-sheet imaging for multimodal validation. Such refinements will broaden the applicability of high-resolution ex vivo diffusion MRI and reinforce its role in preclinical connectomics.

5 | Conclusion

This study represents an optimisation of both the multishot 3D EPI sequence and Gd-DOTA administration for diffusion MRI of the mouse brain ex vivo. When combined with Gd-DOTA administration, the protocol reduced acquisition time by a factor of four while preserving data quality. Reliable diffusion metrics and whole-brain tractograms were obtained after ~11 h of acquisition, compared with ~41 h using conventional long-TR acquisitions. Importantly, our approach, immersing extracted brains in a PBS solution containing Gd-DOTA, also allows the reuse of previously collected samples, offering greater experimental flexibility and reducing animal use. The optimised protocol yields high-quality, high-resolution ex vivo images suitable for tractography and advanced microstructural modelling. Overall, it paves the way for wider adoption of 3D SE-msEPI sequences in preclinical research.

Author Contributions

Elise Cosenza: conceptualisation, data acquisition, data analysis, writing original draft, review and editing. **Nadège Corbin:** data acquisition, data analysis, review and editing. **Emeline J. Ribot:** data acquisition, data analysis, review and editing. **Arnaud Boré:** data analysis. **Maxime Descoteaux:** data analysis, review and editing. **Sylvain Miraux:** conceptualisation, data acquisition, review and editing, supervision, funding acquisition. **Laurent Petit:** conceptualisation, data analysis, writing, review and editing, supervision, funding acquisition.

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Data Availability Statement

The data that support the findings of this study are available from the corresponding author upon reasonable request.

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Supporting Information

Additional supporting information can be found online in the Supporting Information section. **Figure S1:** Summary of sample preparation protocols. **Figure S2:** Simulations of the MR signal obtained from a spin-echo sequence as a function of flip angle and TR.